

SATHISH KUMAR MADIGA MALLAYOLLA

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PROFESSIONAL SUMMARY

Full Stack Developer with 3+ years of experience building scalable, secure, and cloud-native applications using Java, Spring Boot, React, and AWS. Experienced in designing microservices, developing high-performance REST APIs, and deploying containerized applications in production environments. Strong background in database optimization, authentication systems (JWT/OAuth2), and CI/CD pipelines. Additionally skilled in integrating AI/ML capabilities into full stack applications to deliver intelligent, real-time features.

TECHNICAL SKILLS

Languages: Java (8+), Python, JavaScript (ES6+), SQL, PHP

Backend: Spring Boot, Spring MVC, Spring Security, Node.js, Express.js, Hibernate, JPA, RESTful APIs, Microservices Architecture.

Frontend: React.js, Redux, HTML5, CSS3, Responsive Design, Bootstrap, jQuery, AJAX, Cross-Browser Compatibility

Databases: PostgreSQL, MySQL, MongoDB, SQL Server

Cloud & DevOps: AWS (EC2, S3), Docker, Jenkins, CI/CD, Git, GitHub, Maven

Testing: JUnit, Mockito, Jest, Selenium, Postman, API Testing, Unit Testing, Integration Testing

AI & Intelligent Systems: Scikit-learn, NumPy, Pandas, Model Evaluation, Feature Engineering, OpenAI API, Prompt Engineering, RAG, Data Preprocessing, Classification Models

Engineering Principles: OOP, SOLID, Design Patterns, MVC, Data Structures & Algorithms, Agile/Scrum, Secure Coding Practices

PROFESSIONAL EXPERIENCE

Full Stack Developer | YTT Global Services | USA

January 2025 – Present

- Led end-to-end development of scalable full stack applications using React.js, Node.js, and PostgreSQL, supporting secure multi-user workflows in production environments
- Designed and optimized RESTful APIs with efficient database queries, improving response time by 20% and enhancing system performance
- Built reusable and modular UI components using React Hooks and Redux, reducing code redundancy by 30% and improving maintainability
- Implemented secure authentication and authorization using JWT and role-based access control for distributed systems
- Integrated Stripe payment gateway with secure transaction handling and webhook validation for reliable payment processing
- Optimized database performance through indexing and query tuning, improving data retrieval efficiency
- Improved backend reliability by writing unit tests with JUnit and Mockito and validating APIs using Postman, reducing production issues by 20%.
- Deployed containerized applications using Docker and managed CI/CD pipelines for consistent and stable releases

Programmer Analyst | Cognizant Technology Solutions | India

January 2021 – December 2022

- Developed enterprise-grade full stack applications using Java, Spring Boot, React.js, and Node.js in Agile environments
 - Designed and implemented scalable microservices and REST APIs to support secure system integrations
 - Built responsive frontend applications and ensured seamless integration with backend services
 - Implemented Spring Security and JWT authentication to secure APIs and protect sensitive data
 - Improved database performance through optimized SQL queries and Hibernate/JPA configurations
 - Supported production systems by performing debugging, performance tuning, and release management
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PROJECTS

HeartSense AI

- Developed and deployed ML-based prediction service exposed via REST API for real-time inference in web applications.

- Evaluated model performance using cross-validation, ROC-AUC, confusion matrix, and precision-recall metrics.
- Built REST-based backend integration to enable real-time prediction capabilities within web applications.
- Improved model accuracy through iterative hyperparameter tuning and data preprocessing strategies.

AI Service Integration Prototype (Full Stack + ML Integration)

- Built Spring Boot APIs to integrate Python-based ML services for real-time inference.
- Designed frontend UI using React for interactive data submission and prediction visualization.
- Containerized AI services using Docker and deployed on AWS EC2 for scalable access.
- Reduced prediction latency by 30% after API optimization.

EDUCATION

Master's in Computer Science – Western Michigan University

Bachelor's in Computer Science – MGIT

CERTIFICATIONS

- Microsoft SQL Server Development
- Cybersecurity Essentials – Cisco
- CCNAv7 – Cisco
- Advanced Embedded Systems